

# SuperDoc math: m:box / m:borderBox spec coverage test

Each scenario below tests a specific branch of ECMA-376 §22.1.2.11-14 and §22.9.2.7. Labels describe the expected MathML.

## A. m:borderBox defaults

1. No borderBoxPr (expect notation=box)

$abc$

2. Empty borderBoxPr (expect notation=box)

$abc2$

## B. ST\_OnOff value variants (on m:hideTop)

3. m:val=1 (expect top hidden)

$T1$

4. m:val=true (expect top hidden)

$Ttr$

5. m:val=on (Annex L.6.1.3 uses this — expect top hidden)

$Ton$

6. Bare tag hideTop/ (expect top hidden — spec default val=1)

$Tbr$

7. m:val=0 (expect NOT hidden)

$T0$

8. m:val=false (expect NOT hidden)

$Tf$

## C. Individual hide flags in isolation

9. hideBot only (expect top left right)

$\boxed{B}$

10. hideLeft only (expect top bottom right)

$\boxed{L}$

11. hideRight only (expect top bottom left)

$\boxed{R}$

12. hideTop + hideRight — spec §22.1.2.12 example (expect bottom left)

$\boxed{x}$

13. All four hides, no strikes (expect fallback to mrow, no border)

*nobdr*

## D. Strike directions (all borders hidden)

14. strikeBLTR alone (BL to TR = slash = updiagonalstrike)

*BLTR*

15. strikeTLBR alone (TL to BR = backslash = downdiagonalstrike)

*TLBR*

16. strikeH alone (horizontalstrike)

*H*

17. strikeV alone (verticalstrike)

*V*

## E. Strike combinations

18. strikeBLTR + strikeTLBR = X pattern

*X*

19. All four strikes

~~all~~

20. Default box + strikeH (expect: box horizontalstrike)

boxH

21. hideTop + strikeH (expect: bottom left right horizontalstrike)

pH

## F. Spec Annex L.6.1.3 example

22. a squared = b squared + c squared crossed out with TL-BR diagonal, using m:val=on from spec

$a^2 = b^2 + c^2$

## G. m:box scenarios

23. Plain m:box with content == (expect mrow)

==

24. m:box with boxPr/opEmu — spec §22.1.2.13 example (opEmu currently dropped)

$a == b$

25. m:box with boxPr/noBreak (noBreak currently dropped)

$nbr$

26. m:box with all boxPr flags (all currently dropped)

$pAll$

27. Empty m:box (expect null/dropped)

## H. Nested structures

28. m:box inside m:borderBox

$nx$

29. m:borderBox inside m:box

ny

30. Nested m:borderBox inside m:borderBox (outer partial-hide + inner strike)

~~nz SD-2672 plain 3x3~~

|      |      |      |
|------|------|------|
| r0c0 | r0c1 | r0c2 |
|      | r1c1 | r1c2 |
| r1c0 |      |      |
| r2c0 | r2c1 | r2c2 |